

THARIMPEPE FEEDING SCHEME FEEDING SCHEME MANAGEMENT SYSTEM

System Design Document

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Module:

XISD5319 - Work Integrated Learning
Task 2 - System Design

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GitHub:

[ST10345327/fsms---tharimpepe: Feeding Scheme Management System](https://github.com/ST10345327/fsms---tharimpepe: Feeding Scheme Management System)

Client:

Tharimpepe Feeding Scheme



1. Introduction

1.1 Purpose

This System Design Document describes the architectural and detailed design of the Feeding Scheme Management System (FSMS) developed for Tharimpepe Feeding Scheme. It translates requirements into a concrete technical blueprint covering logical architecture, high-level and low-level design, user interaction design, database design, ERD, UI mockups, and report design.

1.2 System Overview

The FSMS is a web-based application that digitalises and streamlines the operations of Tharimpepe Feeding Scheme in Mafikeng, North West, South Africa. It manages beneficiary registration, daily attendance tracking, food stock and donation management, volunteer scheduling, and automated reporting.

1.3 Technology Stack

Layer	Technology	Purpose
Frontend (View)	HTML5, CSS3, JavaScript, Bootstrap 5	User interface – responsive, accessible web pages
Backend (Domain)	PHP 8 (MVC Pattern)	Business logic, routing, session management
Database (Data)	MySQL 8.0 via XAMPP	Persistent data storage for all system entities
Web Server	Apache (via XAMPP)	Serve PHP application locally or on hosting platform
Version Control	Git / GitHub	Source code management and team collaboration
IDE	Visual Studio Code	Development environment (free, cross-platform)
Hosting	AWS Free Tier / cPanel	Cloud deployment of the web application

2. Logical Architectural Design

The FSMS follows a Three-Layer Architecture (View, Domain, Data) that separates concerns for maintainability, scalability, and testability.

2.1 High-Level Architecture Diagram

The diagram below shows all components within each layer and communication pathways between them:

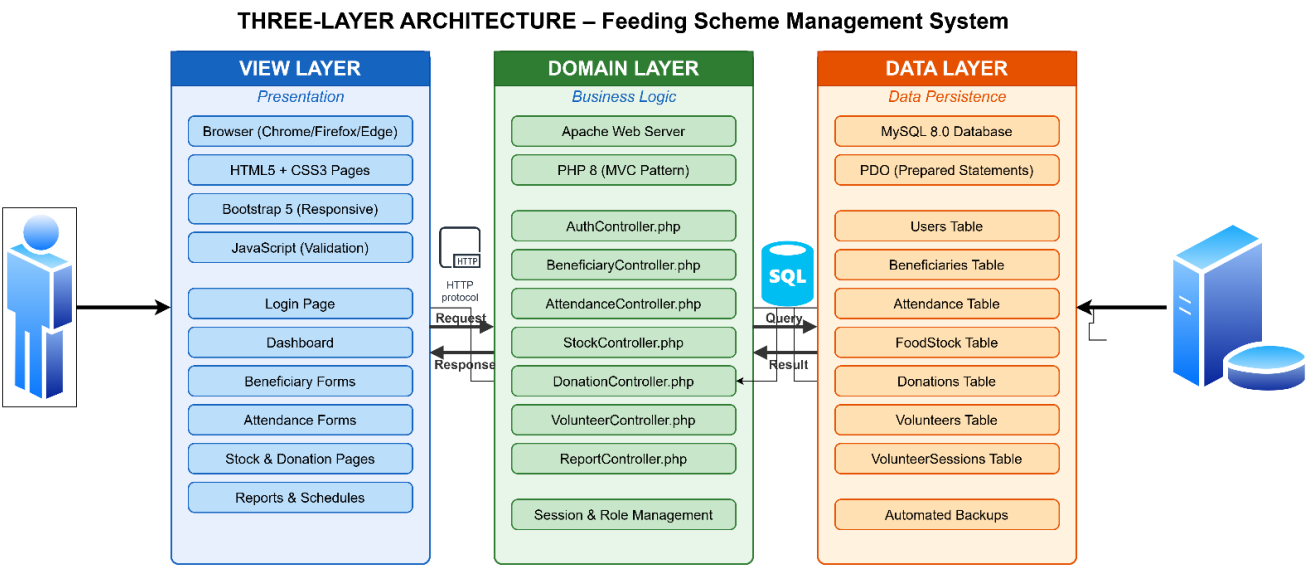


Figure 1: Three-Layer Architecture Diagram – Feeding Scheme Management System

Feeding Scheme Management System High-Level Architecture Diagram

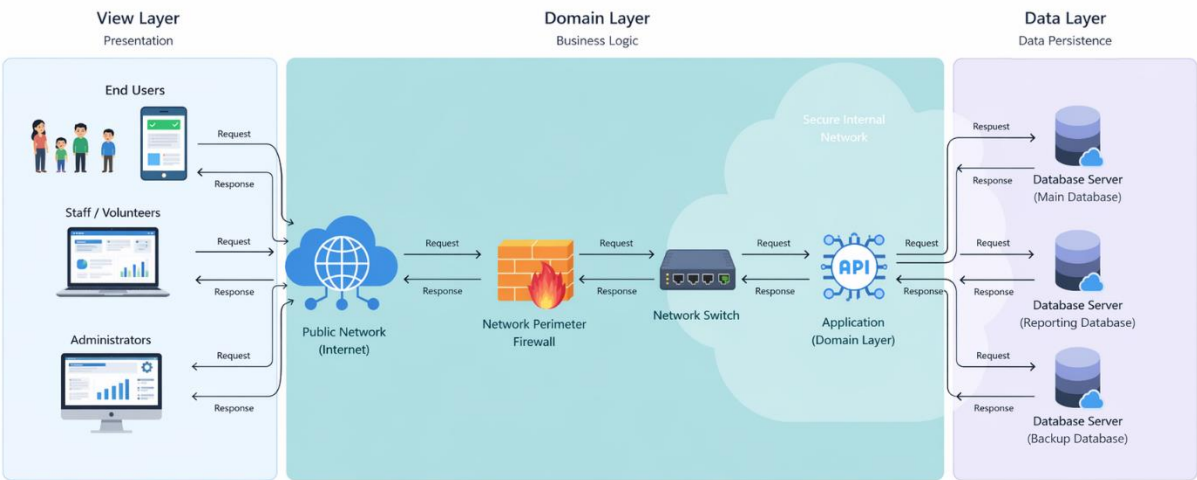


Figure 2: High-Level Architecture Diagram

Layer Descriptions

View Layer: HTML/CSS/JavaScript pages using Bootstrap 5 for responsive, accessible design. Communicates with the domain layer via HTTP GET/POST requests.

Domain Layer: PHP 8 MVC controllers handle business logic, session management, authentication, and input validation before querying or updating the database.

Data Layer: MySQL 8.0 accessed via PDO prepared statements to prevent SQL injection. Seven relational tables store all system data persistently.

2.2 Low-Level Design

2.2.1 MVC Folder Structure

Folder / File	Purpose
/app/controllers/	PHP controller classes (AuthController, BeneficiaryController, AttendanceController, etc.)
/app/models/	PHP model classes (User.php, Beneficiary.php, Attendance.php, FoodStock.php, etc.)
/app/views/	HTML/PHP template files for each page (login.php, dashboard.php, beneficiaries.php, etc.)
/config/	Database connection config (db.php), environment settings
/public/	Entry point index.php + publicly accessible CSS, JS, image assets
/sql/	Database schema files (schema.sql, seed_data.sql)
.htaccess	Apache URL rewrite rules for clean RESTful-style URLs

2.2.2 MVC Request-Response Flow

The diagram below shows the complete seven-step request-response cycle from browser action through controller and model to database and back:

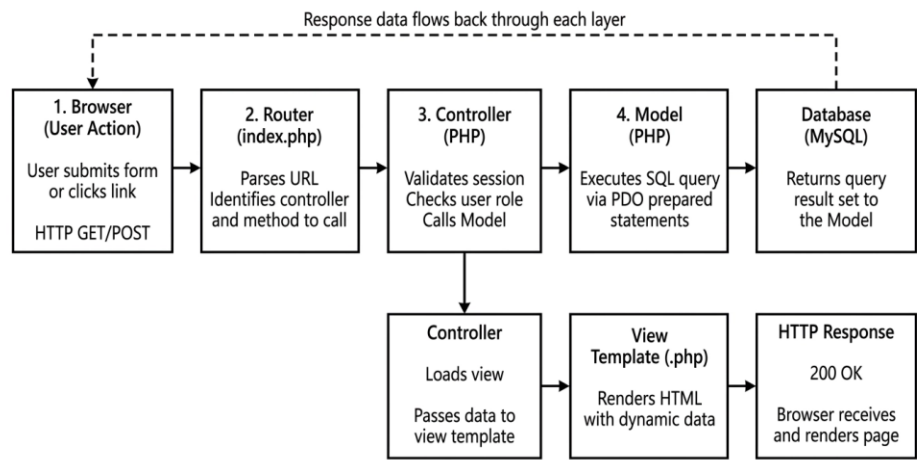


Figure 3: MVC Request-Response Flow Diagram

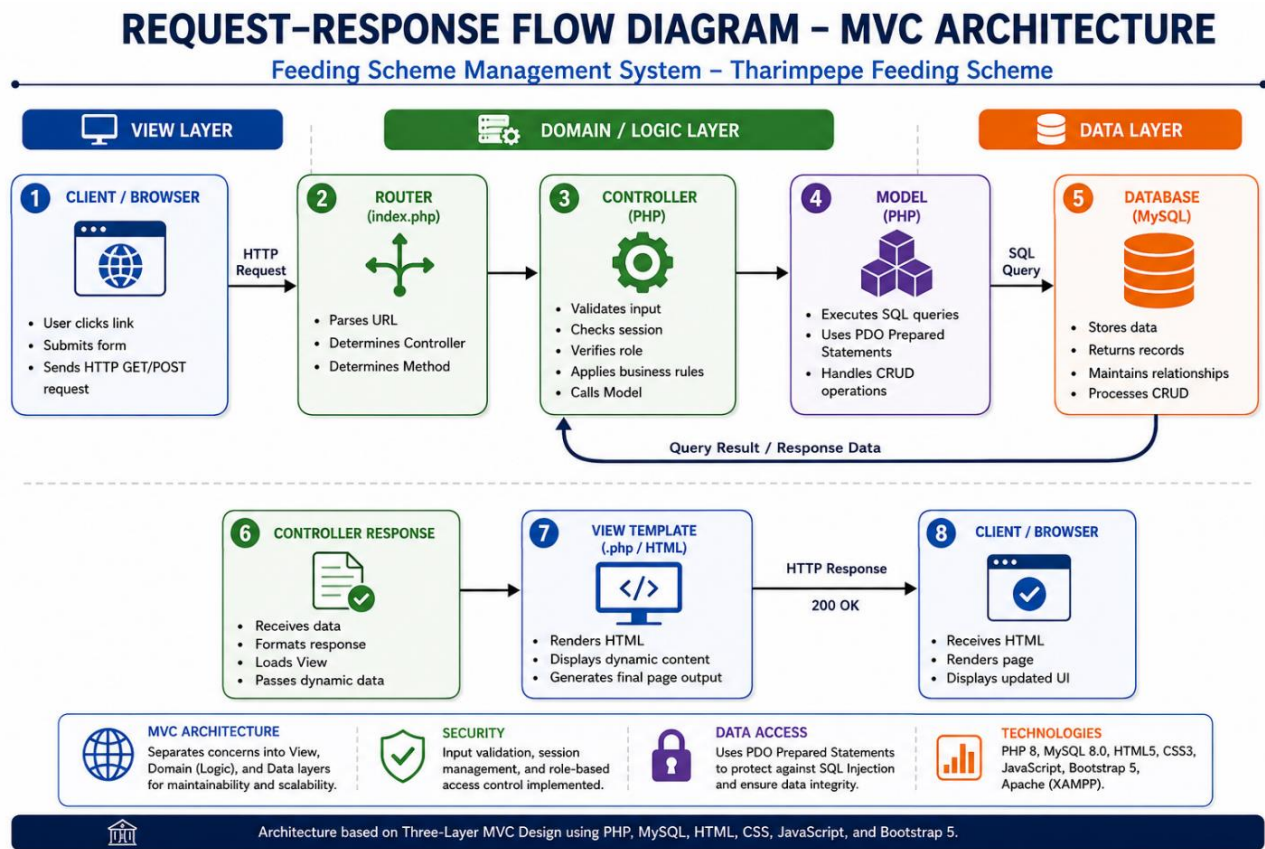


Figure 4: MCV Architecture Diagram

Step	Component	Action
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1	Browser	User submits form or clicks link; browser sends HTTP GET/POST request to the server.
2	index.php (Router)	Front controller parses URL to determine which controller and method to invoke.
3	Controller	Validates session/role; calls appropriate Model method(s) with input data.
4	Model	Executes prepared SQL query against MySQL database; returns result as PHP array.
5	Controller	Receives model result; formats data; loads appropriate view template.
6	View	PHP template renders HTML with dynamic data embedded; HTTP response sent to browser.
7	Browser	Browser renders HTML page to the user.

3. User Interaction Design

3.1 Input Interactions

Screen/Form	Input Fields	Validation	Action on Submit
Login Form	Username, Password	Both required; password min 8 chars	Authenticate user; redirect to dashboard or show error
Register Beneficiary	First Name, Last Name, DOB, Gender, Guardian, Contact, Address	All required except Guardian; DOB must be past date	Save to Beneficiaries table; show success message
Edit Beneficiary	Same fields (pre-filled)	Same as register; at least one field changed	Update Beneficiaries record; redirect to list
Record Attendance	Session Date, Present/Absent per beneficiary	Session date required; at least one beneficiary	Save attendance records; redirect to list
Add Food Stock	Item Name, Category, Quantity, Unit, Min Threshold, Expiry	Quantity > 0; Expiry must be future date	Insert or update FoodStock; show updated inventory
Record Donation	Donor Name, Type, Description, Quantity/Amount, Date	Donor Name, Type, Date required; Amount > 0 if Cash	Insert Donations record; optionally update FoodStock
Register Volunteer	Full Name, Contact, Email, Availability	Full Name and Contact required; valid email format	Insert Volunteers record; show confirmation
Assign Volunteer	Volunteer (dropdown), Session Date, Role	All required	Insert VolunteerSessions record; show schedule
Generate Report	Report Type, Start Date, End Date	Dates required; Start before End	Query database; render report on screen

3.2 Request Interactions (URL Routing)

URL Route	Method	Controller::Method	Description
/login	GET/POST	AuthController::login()	GET: show form. POST: authenticate user.
/logout	GET	AuthController::logout()	Destroy session; redirect to login.
/dashboard	GET	DashboardController::index()	Show dashboard with summary stats.
/beneficiaries	GET	BeneficiaryController::index()	List all active beneficiaries.
/beneficiaries/add	GET/POST	BeneficiaryController::add()	GET: show form. POST: save beneficiary.
/beneficiaries/edit/{id}	GET/POST	BeneficiaryController::edit()	GET: show form. POST: update record.
/attendance/record	GET/POST	AttendanceController::record()	GET: show form. POST: save attendance.
/stock	GET	StockController::index()	View food stock levels.
/stock/add	GET/POST	StockController::add()	GET: show form. POST: save stock.
/donations/add	GET/POST	DonationController::add()	GET: show form. POST: save donation.
/volunteers/add	GET/POST	VolunteerController::add()	GET: show form. POST: save volunteer.
/volunteers/assign	GET/POST	VolunteerController::assign()	GET: show form. POST: assign session.
/reports/generate	POST	ReportController::generate()	Generate and display selected report.

3.3 System Workflows

3.3.1 Beneficiary Registration Workflow

Beneficiary Registration Workflow

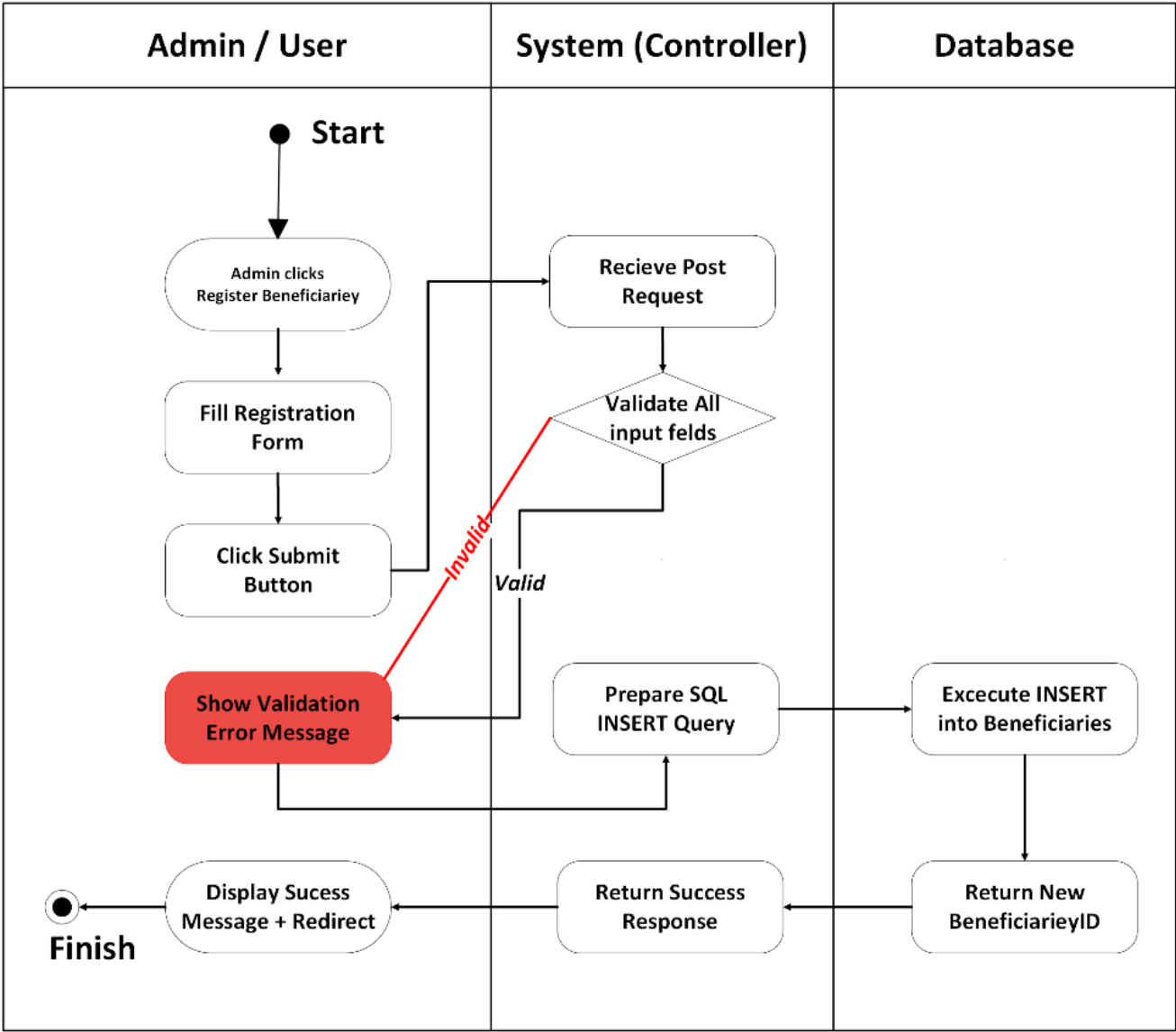


Figure 5:Beneficiary Registration Workflow (Swimlane Diagram)

3.3.2 Daily Attendance Recording Workflow

Daily Attendance Recording Workflow

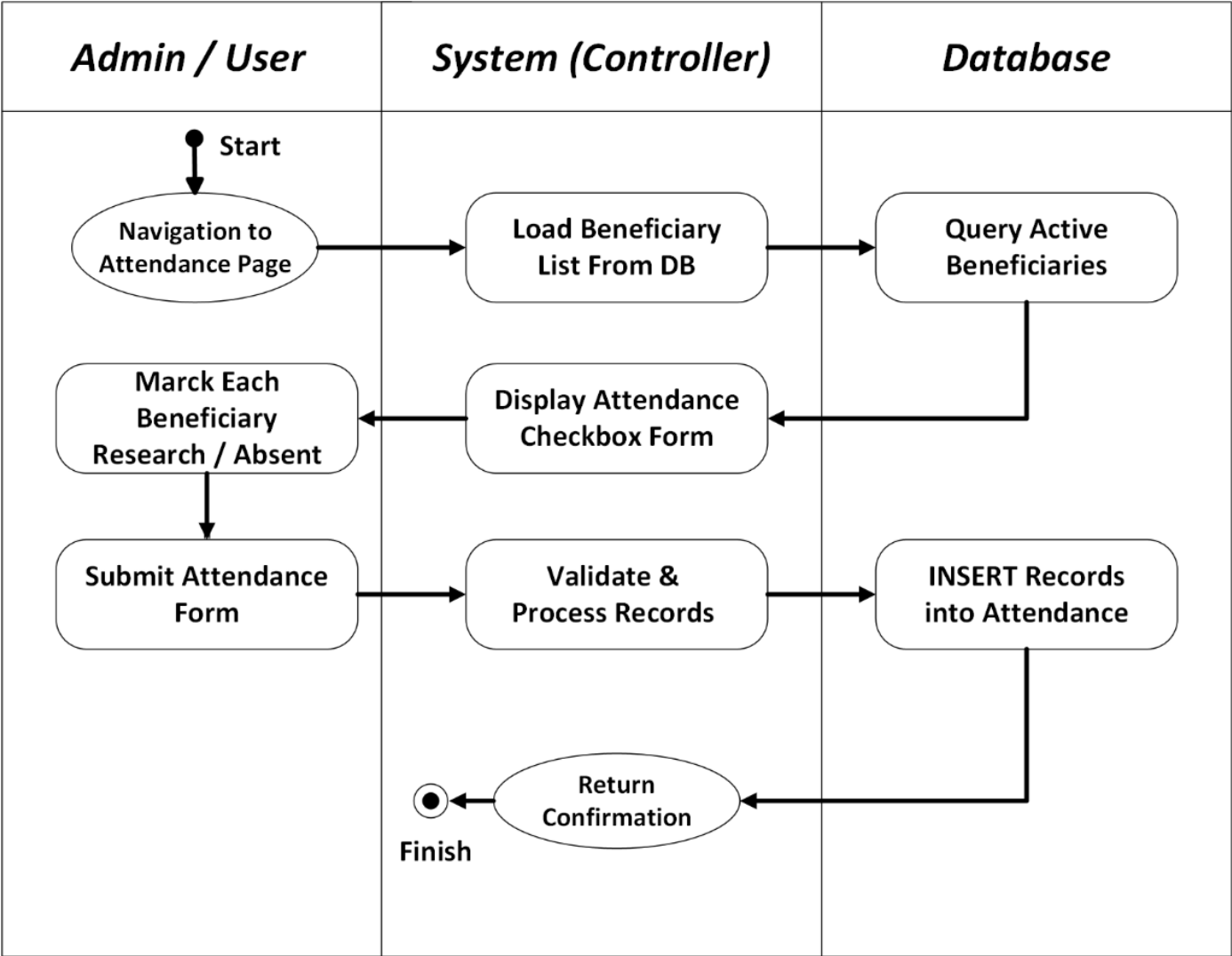


Figure 6: Daily Attendance Recording Workflow (Swimlane Diagram)

3.3.3 Report Generation Workflow

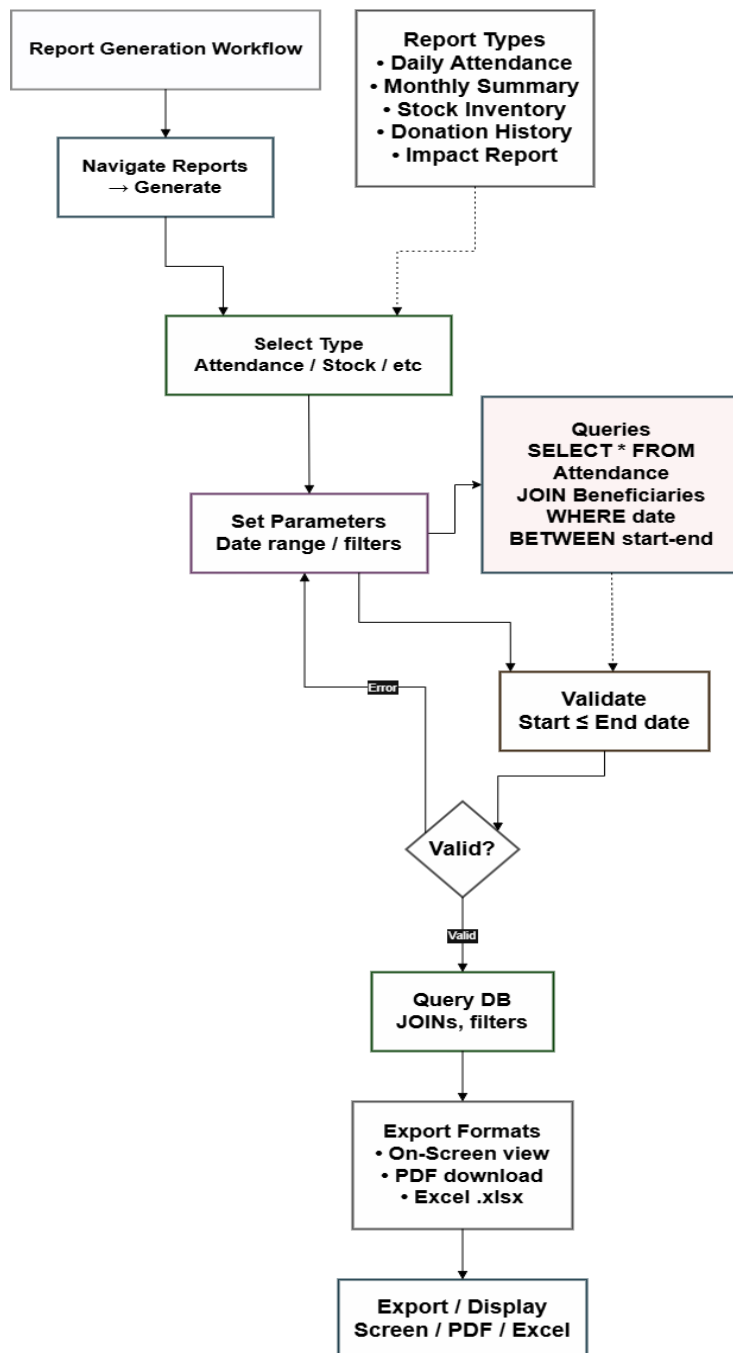


Figure 7: Report Generation Workflow

3.3.4 Donation and Stock Update Workflow

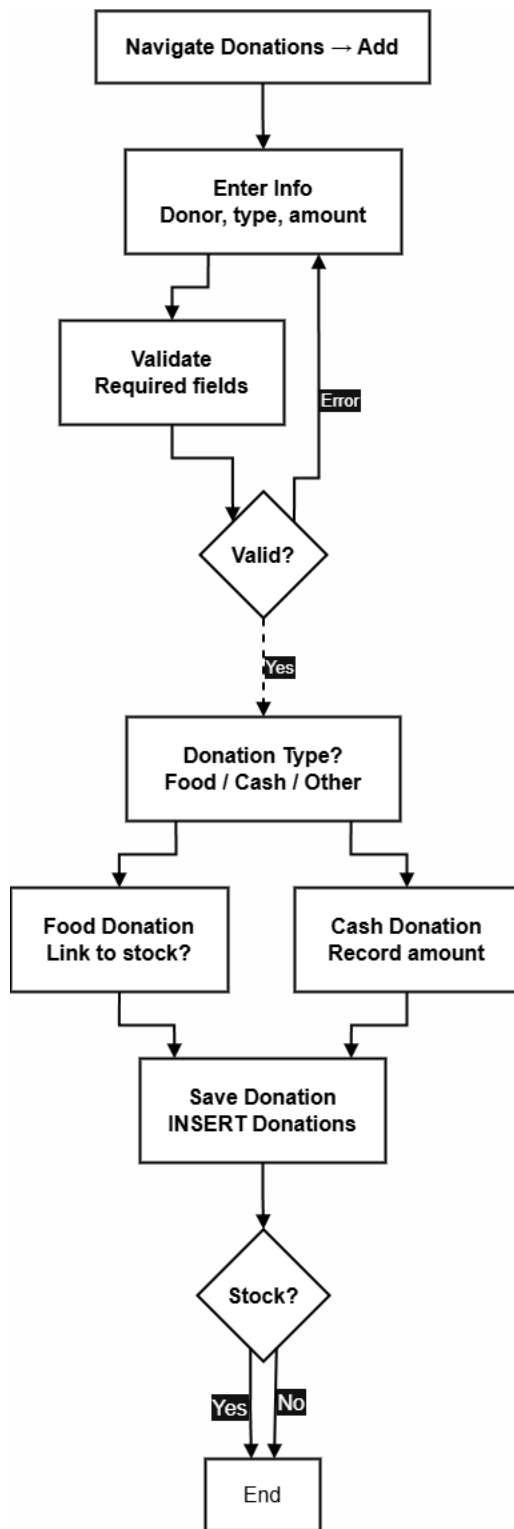


Figure 8: Donation and Stock Update Workflow

4. Database Design

The database is a relational MySQL 8.0 database consisting of seven tables. All tables use InnoDB engine with utf8mb4 character set.

4.1 ERD Design

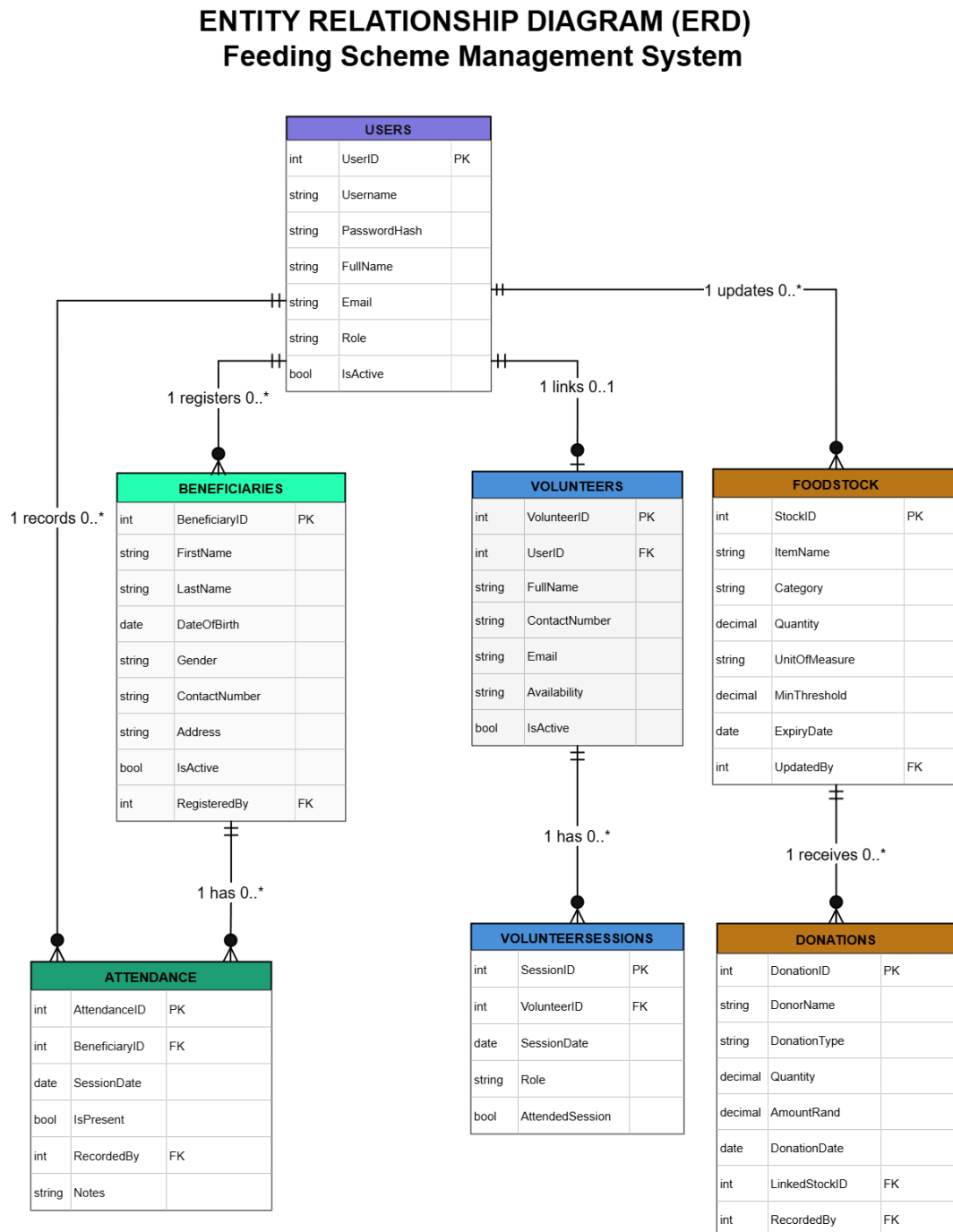


Figure 9: Entity Relationship Diagram (ERD) – MySQL Database Schema

4.2 Database Tables (SQL)

4.2.1 Users

```
CREATE TABLE Users (  
  UserID      INT AUTO_INCREMENT PRIMARY KEY,  
  Username    VARCHAR(50) NOT NULL UNIQUE,  
  PasswordHash VARCHAR(255) NOT NULL,  
  FullName    VARCHAR(100) NOT NULL,  
  Email       VARCHAR(100) NOT NULL UNIQUE,  
  Role        ENUM('Admin','Volunteer') NOT NULL DEFAULT 'Volunteer',  
  IsActive    BOOLEAN NOT NULL DEFAULT TRUE,  
  CreatedAt   DATETIME NOT NULL DEFAULT NOW()  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

4.2.2 Beneficiaries

```
CREATE TABLE Beneficiaries (  
  BeneficiaryID INT AUTO_INCREMENT PRIMARY KEY,  
  FirstName     VARCHAR(50) NOT NULL,  
  LastName      VARCHAR(50) NOT NULL,  
  DateOfBirth   DATE NOT NULL,  
  Gender        ENUM('Male','Female','Other') NOT NULL,  
  GuardianName  VARCHAR(100) NULL,  
  ContactNumber VARCHAR(15) NOT NULL,  
  Address       VARCHAR(200) NOT NULL,  
  RegistrationDate DATE NOT NULL DEFAULT (CURDATE()),  
  IsActive      BOOLEAN NOT NULL DEFAULT TRUE,  
  RegisteredBy  INT NOT NULL,  
  FOREIGN KEY (RegisteredBy) REFERENCES Users(UserID)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

4.2.3 Attendance

```
CREATE TABLE Attendance (  
  AttendanceID INT AUTO_INCREMENT PRIMARY KEY,  
  BeneficiaryID INT NOT NULL,  
  SessionDate  DATE NOT NULL,  
  IsPresent    BOOLEAN NOT NULL,  
  RecordedBy   INT NOT NULL,  
  Notes        TEXT NULL,  
  FOREIGN KEY (BeneficiaryID) REFERENCES Beneficiaries(BeneficiaryID),  
  FOREIGN KEY (RecordedBy) REFERENCES Users(UserID),  
  UNIQUE KEY unique_att (BeneficiaryID, SessionDate)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

4.2.4 FoodStock

```
CREATE TABLE FoodStock (  
  StockID          INT AUTO_INCREMENT PRIMARY KEY,  
  ItemName         VARCHAR(100) NOT NULL,  
  Category         VARCHAR(50) NOT NULL,  
  Quantity         DECIMAL(10,2) NOT NULL DEFAULT 0,  
  UnitOfMeasure    ENUM('kg','litres','units') NOT NULL,  
  MinimumThreshold DECIMAL(10,2) NOT NULL DEFAULT 5,  
  ExpiryDate       DATE NULL,  
  LastUpdated      DATETIME NOT NULL DEFAULT NOW(),  
  UpdatedBy        INT NOT NULL,  
  FOREIGN KEY (UpdatedBy) REFERENCES Users(UserID)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

4.2.5 Donations + Volunteers + VolunteerSessions

```
CREATE TABLE Donations (  
  DonationID      INT AUTO_INCREMENT PRIMARY KEY,  
  DonorName       VARCHAR(100) NOT NULL,  
  DonationType    ENUM('Food','Cash','Other') NOT NULL,  
  Description     VARCHAR(200) NULL,  
  Quantity        DECIMAL(10,2) NULL,  
  AmountRand      DECIMAL(10,2) NULL,  
  DonationDate    DATE NOT NULL DEFAULT (CURDATE()),  
  LinkedStockID   INT NULL,  
  RecordedBy      INT NOT NULL,  
  FOREIGN KEY (LinkedStockID) REFERENCES FoodStock(StockID),  
  FOREIGN KEY (RecordedBy) REFERENCES Users(UserID)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;  
  
CREATE TABLE Volunteers (  
  VolunteerID     INT AUTO_INCREMENT PRIMARY KEY,  
  UserID          INT NULL,  
  FullName        VARCHAR(100) NOT NULL,  
  ContactNumber   VARCHAR(15) NOT NULL,  
  Email           VARCHAR(100) NULL,  
  Availability     TEXT NULL,  
  IsActive        BOOLEAN NOT NULL DEFAULT TRUE,  
  RegisteredDate  DATE NOT NULL DEFAULT (CURDATE()),  
  FOREIGN KEY (UserID) REFERENCES Users(UserID)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;  
  
CREATE TABLE VolunteerSessions (  
  SessionID       INT AUTO_INCREMENT PRIMARY KEY,  
  VolunteerID     INT NOT NULL,  
  SessionDate     DATE NOT NULL,  
  Role            VARCHAR(100) NULL,  
  AttendedSession BOOLEAN NULL DEFAULT NULL,  
  FOREIGN KEY (VolunteerID) REFERENCES Volunteers(VolunteerID)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

5. User Interface Mockups (Wireframes)

The following high-fidelity wireframes illustrate the complete visual design and layout of all key screens in the Feeding Scheme Management System. Each mockup reflects the Bootstrap 5 responsive design, consistent navigation sidebar, role-based access, and the colour scheme aligned with Tharimpepe Feeding Scheme branding.

All screens share a consistent layout: a dark sidebar navigation (left), a topbar with breadcrumbs (top), and a main content area (right). The colour scheme uses deep navy (#1B3A5C) for branding elements, with green/red status indicators following standard UX conventions.

Figma link: <https://trunk-canon-07981658.figma.site>

5.1 Login Page

The login page is the entry point for all users. It features a split-screen layout with branding on the left and a clean authentication form on the right. Access is restricted to registered system users only.

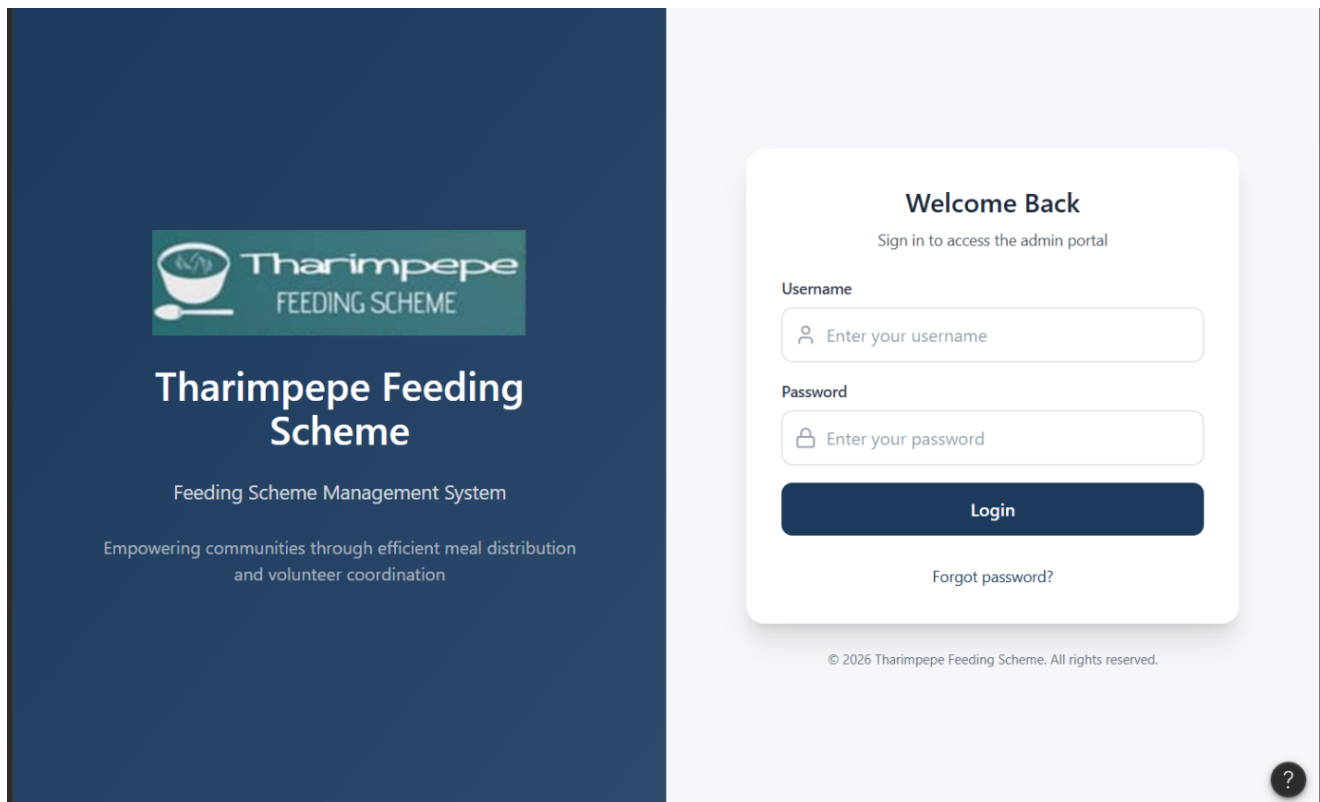


Figure 10: Login Page Mockup

5.2 Admin Dashboard

The dashboard provides an at-a-glance summary of key system metrics: total beneficiaries, meals served today, active volunteers, and stock alerts. It includes a weekly attendance bar chart, food stock progress bars, and a recent activity feed.

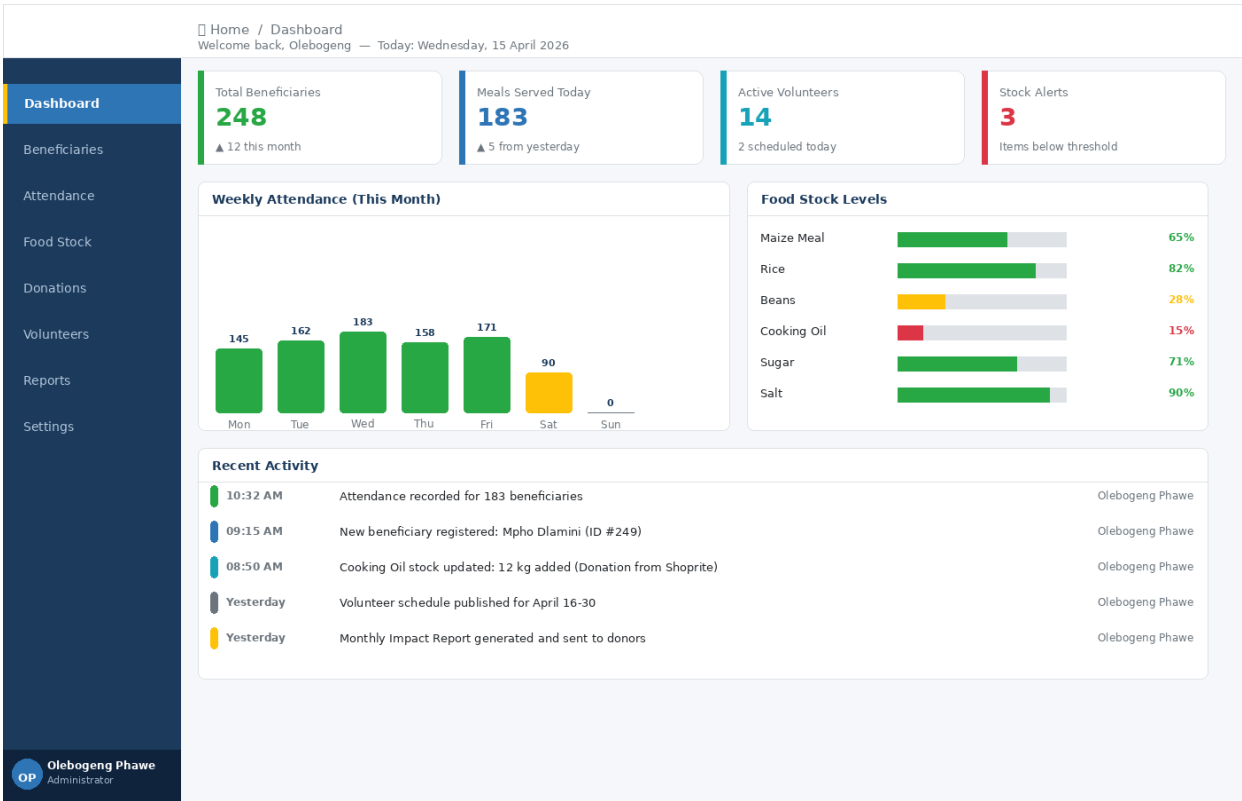


Figure 11: Admin Dashboard Mockup

5.3 Beneficiary List

The beneficiary list displays all registered beneficiaries in a paginated table with search and filter functionality. Each row shows the beneficiary ID, name, date of birth, gender, contact number, registration date, and active status badge. Edit and delete action buttons appear per row.

Home / Beneficiaries

Manage registered beneficiaries

0 Beneficiaries

Showing 248 beneficiaries...

All Status

All Gender

+ Add Beneficiary

ID	Full Name	Date of Birth	Gender	Contact	Registered	Status	Actions
001	Kabo Moagi	12 Jan 2018	Male	072 345 6789	01 Mar 2026	Active	↔ 📄
002	Lerato Sithole	05 Aug 2016	Female	083 456 7890	01 Mar 2026	Active	↔ 📄
003	Thabo Nkosi	22 Nov 2019	Male	071 234 5678	05 Mar 2026	Active	↔ 📄
004	Nomsa Dlamini	18 Apr 2015	Female	082 567 8901	08 Mar 2026	Active	↔ 📄
005	Mpho Tau	30 Jun 2017	Male	084 678 9012	10 Mar 2026	Active	↔ 📄
006	Kagiso Pule	14 Sep 2020	Female	073 789 0123	15 Mar 2026	Inactive	↔ 📄
007	Refilwe Mabe	03 Feb 2018	Female	081 890 1234	20 Mar 2026	Active	↔ 📄
008	Lesego Sello	27 Dec 2016	Male	074 901 2345	22 Mar 2026	Active	↔ 📄
009	Mmabatho Tlou	11 Jul 2019	Female	085 012 3456	28 Mar 2026	Active	↔ 📄
010	Karabo Setlhabi	09 Mar 2021	Male	076 123 4567	01 Apr 2026	Active	↔ 📄

Showing 1-10 of 248 beneficiaries

[«](#)
[1](#)
[2](#)
[3](#)
[...](#)
[25](#)
[»](#)

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Figure 12: Beneficiary List Mockup

5.4 Register New Beneficiary Form

The registration form collects all required personal details for a new beneficiary: name, date of birth, gender, guardian (optional), contact number, address, medical conditions, and dietary needs. Required fields are marked with an asterisk (*). The form validates all inputs before submission.

Home / Beneficiaries / Register New

Complete all required fields (*)

Dashboard

Beneficiaries

Attendance

Food Stock

Donations

Volunteers

Reports

Settings

OP Olebogeng Phawe Administrator

Register New Beneficiary

Personal Information

First Name *
e.g. Kabo

Last Name *
e.g. Moagi

Date of Birth *
DD/MM/YYYY

Gender *
☒ Male ☐ Female ☐ Other

Guardian / Parent Name
Full name of guardian

Contact Number **
e.g. 072 345 6789

Physical Address *
Street address, suburb, city

Additional Information

Medical Conditions / Allergies
Optional

Special Dietary Needs
Optional

Notes

Save Record

Reset Form

Cancel

* Required fields

Figure 13: Register Beneficiary Form Mockup

5.5 Record Daily Attendance

The attendance screen loads the full beneficiary list for a selected session date. Each beneficiary has toggle buttons for Present/Absent. The screen shows a live count of present, absent, and total beneficiaries. Quick-action buttons allow marking all present or all absent at once. Notes can be added per beneficiary.

Home / Attendance

Record daily feeding session attendance

Session Date:

15 April 2026

Load Beneficiaries

Session: Morning Feeding | 183 beneficiaries loaded

Submit All

Present

183

Absent

65

Total

248

Quick actions:

Mark All Present

Mark All Absent

Reset All

ID	Beneficiary Name	DOB	Present	Absent	Notes
1 001	Kabo Moagi	12 Jan 2018	☒	☐	
2 002	Lerato Sithole	05 Aug 2016	☒	☐	
3 003	Thabo Nkosi	22 Nov 2019	☐	☒	Absent - illness reported
4 004	Nomsa Dlamini	18 Apr 2015	☒	☐	
5 005	Mpho Tau	30 Jun 2017	☒	☐	
6 006	Kagiso Pule	14 Sep 2020	☐	☒	Not at home
7 007	Refilwe Mabe	03 Feb 2018	☒	☐	
8 008	Lesego Sello	27 Dec 2016	☒	☐	
9 009	Mmabatho Tlou	11 Jul 2019	☒	☐	Late arrival
10 010	Karabo Setlhabi	09 Mar 2021	☒	☐	
11 011	Boitumelo Kgosi	15 May 2017	☐	☒	
12 012	Tshegofatso More	22 Oct 2016	☒	☐	

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Figure 14: Record Daily Attendance Mockup

5.6 Food Stock Management

The food stock screen shows current inventory levels in a table with colour-coded status indicators. Items below minimum threshold are highlighted in red with a LOW STOCK badge and a warning banner at the top. Each row includes edit and update-quantity action buttons.

Home / Food Stock

Manage food inventory and stock levels

Search items...

Category ▼

Status ▼

+ Add Stock Item

+ Record Donation

⚠ 3 items are below minimum stock threshold and require urgent restocking.

Item Name	Category	Quantity	Unit	Min. Threshold	Expiry Date	Status	Last Updated	Actions
Maize Meal	Grain	65.0	kg	20.0	30 Jun 2026	OK	Today	⬅️ ➕
White Rice	Grain	42.5	kg	15.0	31 Aug 2026	OK	Today	⬅️ ➕
Red Beans	Protein	8.2	kg	10.0	15 May 2026	LOW	Today	⬅️ ➕
Cooking Oil	Fats	3.5	litres	5.0	28 Feb 2027	LOW	Today	⬅️ ➕
White Sugar	Sweetener	28.0	kg	10.0	31 Dec 2026	OK	Today	⬅️ ➕
Iodised Salt	Condiment	12.0	kg	5.0	31 Mar 2027	OK	Today	⬅️ ➕
Samp	Grain	4.0	kg	8.0	30 Sep 2026	LOW	Today	⬅️ ➕
Sunflower Oil	Fats	18.5	litres	5.0	15 Jan 2027	OK	Today	⬅️ ➕
Bread Flour	Grain	55.0	kg	15.0	20 Jul 2026	OK	Today	⬅️ ➕
Powdered Milk	Dairy	22.0	kg	8.0	28 Feb 2027	OK	Today	⬅️ ➕

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Figure 15: Food Stock Management Mockup

5.7 Volunteer Schedule

The volunteer schedule is presented as a weekly calendar view showing which volunteers are assigned to each day's feeding session. Each assigned slot displays the time and role. Tabs allow switching between the volunteer list, schedule, assignment form, and registration form.

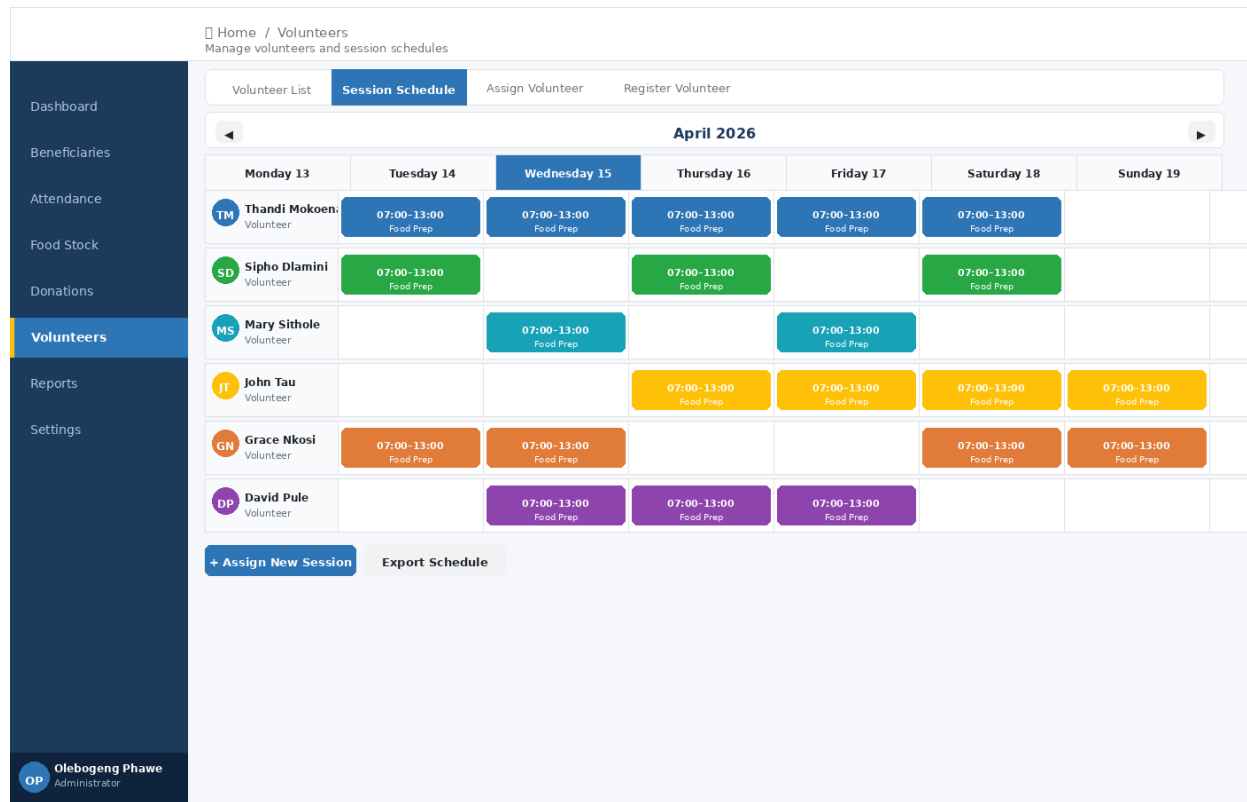


Figure 16: Volunteer Schedule Mockup

5.8 Reports Page

The reports page allows the admin to select a report type, specify a date range and output format (PDF, Excel, or on-screen), and generate the report. A live preview panel shows a formatted summary on the right. Export buttons allow downloading as PDF, Excel, or printing.

Home / Reports

Generate and export reports for stakeholders

Dashboard

Beneficiaries

Attendance

Food Stock

Donations

Volunteers

Reports

Settings

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Daily Attendance Report
View attendance per session

Monthly Attendance Summary
Meals served trend analysis

Food Stock & Inventory
Current stock levels

Donation History Report
All donations by period

Organisational Impact Report
Full stakeholder report

Generate Report

Report Type

Monthly Attendance Summary ▼

Start Date *

01 March 2026

End Date *

31 March 2026

Audience

☐ Admin Internal

☒ Donor Report

☐ Community Report

Export Format

☒ PDF

☐ Excel

☐ On Screen

► Generate Report

Schedule Report

March 2026 — Quick Stats

Total Meals

4,891

Avg Daily

157.8

Top Day

183

Report Preview

THARIMPEPE FEEDING SCHEME
Monthly Attendance Summary Report
Period: 01 March 2026 - 31 March 2026

Date	Present	Absent	Total
01 Mar	152	96	248
05 Mar	168	80	248
10 Mar	183	65	248
15 Mar	145	103	248
20 Mar	171	77	248
25 Mar	158	90	248
31 Mar	162	86	248
TOTAL	4,891		

PDF

Excel

Print

Figure 17: Reports Generation Mockup

6. Report Design

The FSMS generates five reports. All display on-screen and are available for PDF/Excel export.

6.1 Daily Attendance Report

Field	Details
Purpose	Shows beneficiary attendance/absence for a specific feeding session date
Parameters	Session Date (required)
Data Source	Attendance JOIN Beneficiaries WHERE SessionDate = [date]
Output Fields	Beneficiary ID, Full Name, DOB, Gender, Present (Yes/No), Notes
Summary	Total Present: [n] Total Absent: [n] Total: [n]
Format	Table on screen; PDF export
Access	Admin, Volunteer

6.2 Monthly Attendance Summary

Field	Details
Purpose	Summarises meals served and attendance rates per month for donor reporting
Parameters	Start Date, End Date
Data Source	Attendance GROUP BY SessionDate, COUNT(IsPresent = TRUE)
Output Fields	Date, Total Present, Sessions, Average Attendance (%)
Summary	Total Meals: [n] Avg Daily Attendance: [n]
Format	Table + bar chart; PDF export
Access	Admin only

6.3 Food Stock & Inventory Report

Field	Details
Purpose	Shows current stock levels, highlighting items below minimum threshold
Parameters	None; optional Category filter
Output Fields	Item, Category, Quantity, Unit, Min Threshold, Expiry, Status (OK/LOW)
Highlight	LOW STOCK rows shown in red
Format	Table; PDF export
Access	Admin, Volunteer (view)

6.4 Donation History Report

Field	Details
Purpose	Lists all donations received within a date range
Parameters	Start Date, End Date; optional Type filter
Output Fields	Donation ID, Donor, Type, Description, Qty/Amount, Date, Recorded By
Summary	Total Food: [n] Total Cash: R[amount]
Format	Table; PDF export
Access	Admin only

6.5 Organisational Impact Report

Field	Details
Purpose	High-level summary report for donors and stakeholders
Parameters	Period (Month / Quarter / Custom range)
Output Fields	Total Beneficiaries, Meals Served, Volunteers, Donations Received, Stock Status
Format	Summary dashboard + PDF for sharing with donors
Access	Admin only

7. Appendix

7.1 UML Class Diagram

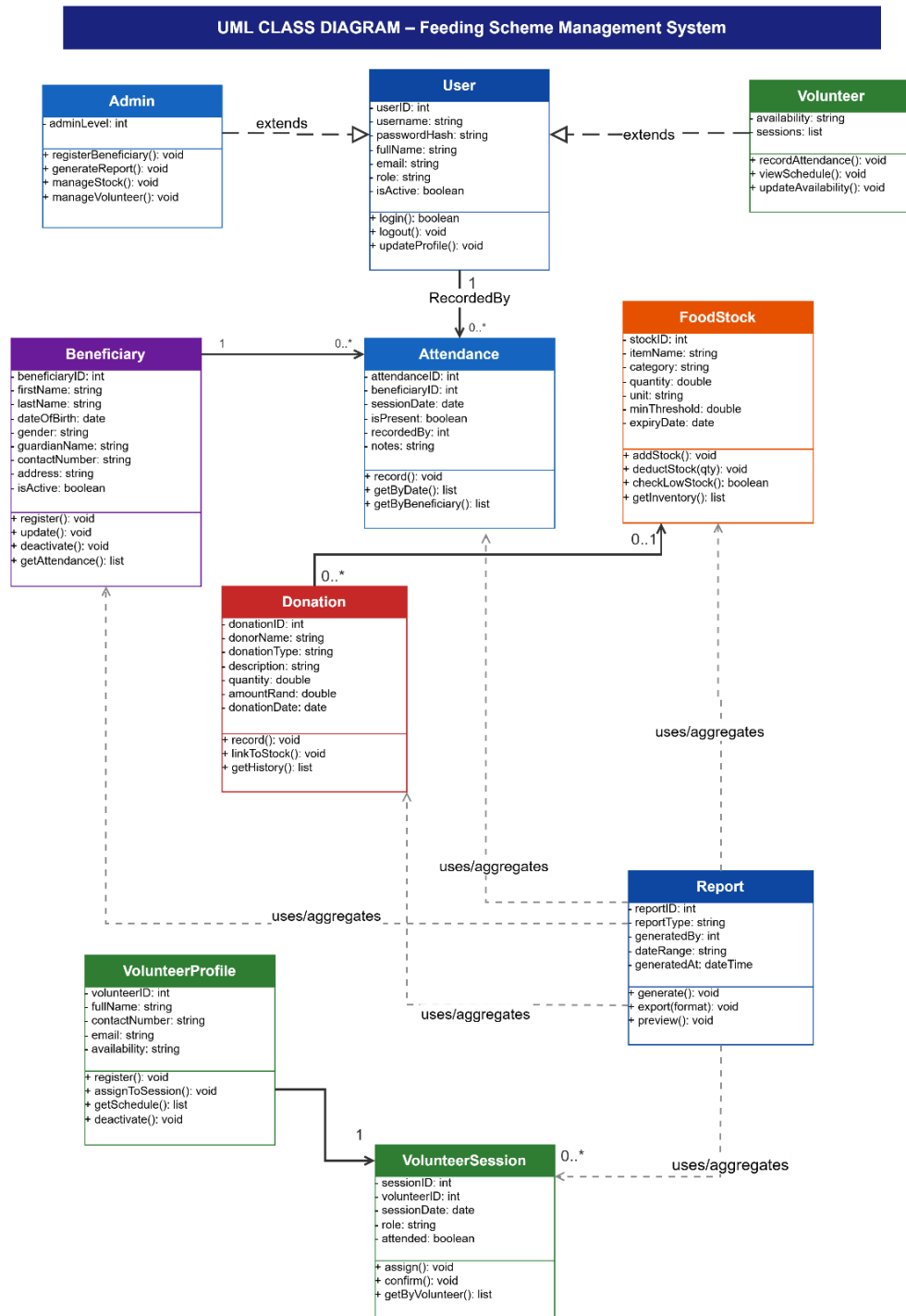


Figure 18: UML Class Diagram – Feeding Scheme Management System

7.2 Use Case Diagram



Figure 19: Use Case Diagram

8. Reference List

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Sources Links

Figma Prototype: <https://trunk-canon-07981658.figma.site>

Github: [ST10345327/fsms---tharimpepe](https://github.com/ST10345327/fsms---tharimpepe): Feeding Scheme Management System